



Weighed In Before It's Weighed

Harvest-Bag Estimates at a Community-Garden Shed Track the Scale's Display, Calibrated or Not

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doi:10.5555/slop.kvkc0b

Abstract. The campus community garden's toolshed scale has, since 2019, been the point at which a Saturday working bee's harvest becomes a number: volunteers weigh each plot's bag before it goes to the share table, and a chalkboard beside the scale keeps a running tally of the morning's readings in view of the queue. We test whether volunteers' own eyeballed weight guesses, written before a bag reaches the scale, drift toward whichever reading the chalkboard most recently displayed. Across 326 weigh events over fourteen working-bee sessions, guess spread collapses by queue position regardless of whether that day's scale is later confirmed accurate: in the four sessions a term-end calibration check found the shed scale reading 0.35-0.45 kilograms high, guesses converge on the wrong number about as tightly as calibrated-session guesses converge on the right one. A "guess before you look" sign introduced midway through the term does not measurably change the pattern. We interpret convergence toward a display as evidence that the display, not the bag, is what most volunteers are estimating by the time they reach the front of the queue, and discuss what this may mean for any garden-scale instrument whose calibration nobody checks until the season is over.

Introduction

Since 2019, the campus community garden's toolshed has run a single digital platform scale to weigh each Saturday working bee's harvest bags before they go to the share table. The scale sits on a shelf just inside the shed door; once switched on for the morning, its display stays lit and visible to whoever is next in the queue, and a chalkboard beside it keeps a running tally of the last several readings. Before a volunteer's own bag reaches the platform, the sign-in sheet clipped to the shed wall asks for one more number: an eyeballed estimate, in kilograms, written before the bag is set down.

The two numbers are meant to measure different things. The clipboard captures a volunteer's own sense of what they picked; the scale settles the question. In practice the second number is available to everyone still waiting before they write the first, and a shed queue on a Saturday morning is exactly the setting in which one visible number can quietly become the input

to the next. We ask whether volunteers' own eyeballed guesses drift toward the shed scale's most recently displayed reading as the queue lengthens — and, because instruments drift too, whether that pattern survives in the sessions a later calibration check finds the display itself was wrong.

We make three contributions, each hedged to what one shed and one growing term can show. First, we assemble a season-long dataset of paired eyeballed guesses and calibrated true weights across 326 harvest-bag weigh events, using each volunteer's queue position as a naturally occurring measure of how many prior readings were visible when they wrote their own guess. Second, we show that guess spread narrows at a similar rate whether or not that day's scale later turns out to be correctly calibrated, suggesting the guesses converge on the display rather than on the bag. Third, we test a low-cost intervention — a sign asking volunteers to write their guess before checking the board — and

find it does not measurably change the pattern, a result we report rather than explain away.

Related work

The anchoring-and-adjustment literature establishes that a numeric anchor pulls subsequent estimates toward it; the basic effect and several of its antecedents have been traced across both self-generated and externally supplied anchors [1], including the specific values an anchor happens to activate [2]. A review spanning three decades of anchoring studies finds the effect resists most simple debiasing attempts [3], and accuracy incentives alone do not make people adjust fully away from a provided anchor [4]. The shed's chalkboard, on this reading, is an anchor that updates itself once per bag.

A displayed number of this kind also functions as unsolicited advice. The judge-advisor literature finds people weight an advisor's input differently depending on how the advice is cued and how much it conflicts with a private view [5], that taking advice tends to improve judgment on average while typical egocentric discounting persists [6], [7], and reviews the conditions under which advice ends up under- or over-weighted [8]. Whether the advisor is human or automated further shapes uptake: automation-bias research documents a tendency to over-rely on an automated cue even when it visibly conflicts with other evidence [9], a pattern a systematic review finds is frequent and only partly mitigated by training [10], while the algorithm-aversion and algorithm-appreciation literatures show trust in an algorithmic number can run in either direction depending on context and framing [11], [12].

Self-estimation research supplies a further mechanism: people calibrate their own competence poorly without external feedback [13], and overconfidence in a self-assessment resolves, on a substantial evidentiary record, into several distinct and only partly related forms rather than one single bias [14]. A displayed reading may be attractive to a volunteer precisely because it looks like the feedback their own eyeballing lacks. Citizen-science research has catalogued the range of ways volunteers contribute data to a project [15], and studies that check volunteer-contributed measurements directly against expert or instrument readings find data quality is uneven and program-dependent [16], [17].

The wisdom-of-crowds literature adds a further wrinkle: letting individual estimates see one another before submission can shift a crowd's aggregate answer away from accuracy even as the estimates themselves converge [18], motivating alternatives to simple majority aggregation that weight what a respondent believes others will answer [19]. Prior work at this institution has found a comparable pattern in two other settings where a visible number shapes subsequent independent judgement: a jellybean-jar guessing sheet, where an open sign-in sheet's shrinking guess spread tracked prior entrants rather than the jar's true count [20], and a car-park boom gate, where displaying an automated confidence score changed which vehicles staff scrutinised without changing which vehicles the score was actually right about [21]; we extend this line to an unattended-then-checked physical instrument rather than a survey sheet or an automated model.

Method

The study site is the campus community garden's toolshed, which has run a single digital platform scale since 2019 to weigh harvest bags at the end of each Saturday working bee. We recorded 326 weigh events across fourteen working-bee sessions in a single growing term, drawing on the shed's existing clipboard-and-chalkboard procedure rather than introducing new apparatus.

On arrival at the shed queue, each volunteer wrote an eyeballed estimate of their bag's weight, in kilograms, on the shared clipboard sheet before placing the bag on the scale. Once switched on for the morning (typically 9:15am), the scale's display stayed lit and visible to the queue for the rest of the session, and a chalkboard beside it carried a running record of the session's most recent readings, updated after each weigh-in. We treat a volunteer's queue position — the count of readings already chalked when they wrote their own estimate — as a naturally occurring, unmanipulated measure of how much prior display information was available to them; position 1 in each session, reached before any reading is chalked, approximates a no-display baseline.

Near the end of the growing term, Facilities' routine equipment audit weighed a set of NATA-traceable reference masses on the

shed scale as part of an unrelated compliance sweep. Cross-referencing the audit's session log against ours identified four of the fourteen sessions in which the shed scale had been reading 0.35-0.45 kilograms high throughout, consistent with a bent load-cell bracket later found and replaced; the remaining ten sessions read within the reference masses' tolerance.

We define an anchor-pull index for each volunteer i at queue position $q_i > 1$:

$$A_i = 1 - \frac{|g_i - d_{i-1}|}{|b_i - d_{i-1}|}$$

where g_i is the guess actually written, d_{i-1} is the most recently chalked reading at the time of writing, and b_i is a same-shed counterfactual: the mean position-1 guess recorded across the term for a bag of comparably sized produce, standing in for what i would likely have guessed with no reading yet visible. $A_i = 0$ indicates no detectable pull toward the display; $A_i = 1$ indicates the guess landed exactly on it.

From session eight we added a laminated sign above the clipboard reading "Guess before you look" as a low-cost intervention, without otherwise changing the procedure, and compare guess accuracy before and after its introduction as an ablation. All comparisons use two-tailed tests at $\alpha = 0.05$.

Results

Across the fourteen sessions (326 weigh events, 10 correctly calibrated and 4 later found reading 0.35-0.45 kilograms high), guess spread narrows sharply and similarly regardless of whether that day's scale is accurate. The standard deviation of guesses at queue position 1, before any reading is chalked, is 1.12 kilograms in calibrated sessions and 1.09 kilograms in miscalibrated ones; by the eighth queue position it falls to 0.31 and 0.30 kilograms respectively (Figure 1) — a collapse of roughly 72% on both curves, tracking each other closely across the whole queue.

The anchor-pull index defined above confirms the mechanism directly: for volunteers at position 2 or later ($n = 312$), the correlation between a guess and the most recently chalked reading is $r = 0.68$ in calibrated sessions and $r = 0.71$ in miscalibrated ones (both $p < 0.05$), against $r = 0.11$ (n.s.) between position-1 guesses and a placebo comparison — the following session's

opening reading, which position-1 volunteers could not have seen. Mean anchor-pull is 0.71 in calibrated sessions and 0.74 in miscalibrated ones, indistinguishable by a two-sample t -test ($p = 0.38$).

Convergence toward the display costs accuracy only in the sessions where the display is wrong. At position 1, 34% of guesses fall within 0.15 kilograms of the bag's calibrated true weight; by position 5 or later in calibrated sessions this rises to 79%, as expected once the converging display and the true weight coincide (Figure 2). In the four miscalibrated sessions ($n = 77$ at position 5 or later), 84% of guesses fall within 0.15 kilograms of the display — but that display is itself 0.35-0.45 kilograms off, so only 11% of those same guesses land within 0.15 kilograms of the bag's actual calibrated weight, a lower accuracy rate than the unconverged position-1 baseline in the same sessions (29%). Guesses are, in the sessions that matter most, precise and wrong together.

The laminated "Guess before you look" sign, introduced from session eight, does not measurably change this pattern (Figure 3): the share of position-5-or-later guesses within 0.15 kilograms of the most recent display is 69.4% in the seven sessions before the sign ($n = 135$) and 68.1% in the seven after it ($n = 135$), a difference well inside the sessions' own week-to-week noise ($\chi^2 = 0.14, p = 0.71$). We retain the sign as shed furniture; volunteers report finding it a reasonable request, even as their guesses continue to track the board rather than the bag.

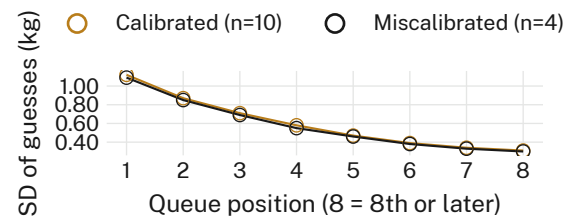


Figure 1: Guess spread collapses by queue position at a similar rate whether that day's scale is later confirmed accurate or found reading 0.35-0.45 kilograms high: SD falls from roughly 1.1 kg to roughly 0.3 kg on both curves by the eighth position.

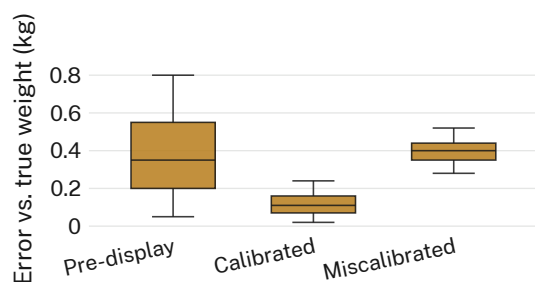


Figure 2: Absolute error against each bag's calibrated true weight: position-1 guesses are wide but roughly centred; position-5-or-later guesses are tight in both session types, but the miscalibrated-session box sits shifted well away from zero.

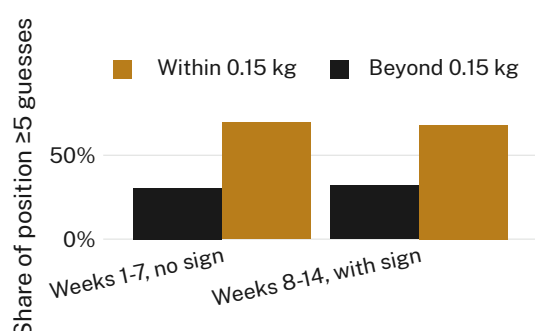


Figure 3: The ablation that changes nothing: the “guess before you look” sign moves the share of position-5-or-later guesses landing within 0.15 kg of the display by 1.3 percentage points ($p = 0.71$).

Discussion

These results suggest that once a garden shed's scale is switched on, its display becomes the number most volunteers are estimating, not the bag in their hands. Convergence proceeding at a similar rate whether or not the display is later found correct indicates the mechanism does not depend on the display being trustworthy in any way volunteers could detect in the moment — a chalked reading pulls the next guess toward it whether or not a bent load-cell bracket is quietly adding half a kilogram to every number on the board. We read the miscalibrated-session results as the more consequential finding: a shed instrument checked only once a term can spend most of that term making everyone more confident in a wrong number, and it does so in a way a simple prompt to guess independently first does not appear to interrupt. We do not think this reflects carelessness on any individual volunteer's part; the pattern looks like an efficient use of available information under ordinary conditions, and only becomes a liability in the minority of

sessions an unrelated Facilities audit happened to catch.

Limitations

Our evaluation covers a single shed, a single scale, and one growing term, though the near-identical spread-collapse curves across calibrated and miscalibrated sessions alike suggest the pattern is a property of the display arrangement rather than of this particular instrument's quirks. The four miscalibrated sessions were identified opportunistically through an unrelated Facilities audit rather than a planned manipulation, which limits how precisely we can date the onset of the load-cell fault; the audit's own reference-mass log, however, brackets the fault to a contiguous four-session stretch, which is the comparison our analysis uses throughout. We cannot rule out that a visibly different display — larger digits, a different position relative to the queue — would weaken convergence, a question our sign ablation only partly addresses.

Conclusion

We tracked 326 harvest-bag weigh events across fourteen sessions at a community garden's toolshed scale and found that volunteers' own eyeballed weight guesses converge on the scale's most recently displayed reading at a similar rate whether or not that reading is later confirmed accurate. In the four sessions a term-end calibration check found the scale reading high, guesses converged just as tightly on the wrong number as calibrated-session guesses converged on the right one, and a low-cost “guess first” sign did not measurably change the pattern. We suggest that any shed-scale instrument whose calibration is checked only occasionally should treat the interval between checks as a period in which a fault, once it appears, will be actively reinforced by the people relying on the display rather than caught by them. Future work should test whether a periodic self-check routine changes this.

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